

Reference-Free Per-Mask Reliability Assessment for Whole-Body Anatomical Segmentation Across CT and MRI

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Abstract

Segmentation tools such as TotalSegmentator increasingly serve as black-box imaging tools inside automated clinical AI pipelines, yet return masks with no per-output reliability signal. On 57 held-out CT subjects, 25.5% of masks fall below an IoU of 0.90 with per-structure pass rates spanning 2.7% to 100%; on an in-sample MRI subset, 77.0% fall below the same bar. The failures are largely silent: of masks failing on overlap, 52.8% (CT) and 69.5% (MRI) pass a naive volume check. We present a reference-free reliability layer that predicts per-mask quality from nineteen geometric and intensity descriptors computed from the predicted mask and source image alone, with no reference annotation, auxiliary network, or segmenter access. It reaches ROC-AUC 0.871 (95% CI 0.852–0.892) on held-out CT and 0.919 (0.890–0.947) on the in-sample MRI subset, improving over organ-identity baselines by 0.110 and 0.104 AUC points, respectively, and transfers to anatomical families withheld from training (mean AUC 0.779). We show the fixed IoU threshold encodes a size-dependent physical standard and replace classification with IoU regression so the operating point is chosen at deployment. Scores are calibrated where data permit (ECE 0.029 to 0.013 on CT) and equipped with conformal intervals (coverage 52.0% to 80.7% on CT, 45.4% to 78.6% on MRI, at nominal 80%), so reported confidence can be acted upon.

Keywords: Medical image segmentation, segmentation quality control, failure detection, uncertainty quantification, conformal prediction, model calibration, CT, MRI

Data and Code Availability We use the publicly available TotalSegmentator CT and MRI datasets (Wasserthal, 2023; Wasserthal and Akinci D’Antonoli, 2025). The datasets are available to researchers through Zenodo. Code is not publicly available at submission and will be released after acceptance.

Institutional Review Board (IRB) This study uses publicly available, de-identified datasets and involves no new human-subjects data collection. IRB approval was therefore not required.

1. Introduction

Quantifying anatomy from medical images supports diagnosis, treatment planning, and monitoring. The TotalSegmentator family segments 104 structures on CT in its original release, 117 currently, and 50 on MRI (Wasserthal et al., 2023a,b; Akinci D’Antonoli et al., 2025), and is widely used as a black-box package in research and clinical pipelines (Sebro and Mongan, 2023). An incorrect mask can therefore produce an incorrect measurement, while verifying over a hundred masks per scan by hand would restore the bottleneck that automation removed. Per-output quality assessment is an important complement to automatic segmentation (Kohlberger et al., 2012; Valindria et al., 2017; Robinson et al., 2019), especially when segmentation is called by automated clinical AI systems and downstream software consumes its measurements without inspecting the mask.

Aggregate accuracy can hide poorly segmented structures. On 57 CT validation subjects held out from the segmenter’s training, 25.5% of masks fall below IoU 0.90, with per-structure pass rates from 2.7% (portal and splenic veins) to 100% (several bones); on a 38-subject MRI subset, in-sample for the segmenter and therefore optimistic, 77.0% fall below the bar. Among masks failing on overlap, 52.8% (CT) and 69.5% (MRI) have relative volume error under 10%, so a volumetric check clears most spatially wrong masks. A fixed overlap threshold also represents different physical errors across structures. For a sphere of radius r with uniform boundary error δ , $\text{IoU} \approx 1 - 3\delta/r$: IoU 0.90 tolerates 2.3 mm on a median liver (1,390 mL) but 0.35 mm on a median adrenal (4.6 mL), or 1.5 versus 0.23 CT voxels; for the adrenal this is roughly one eighth of a voxel at MRI’s coarser spacing.

We propose a lightweight layer that predicts per-mask quality from nineteen descriptors computed from the predicted mask and image alone, without segmenter internals or a reference at inference. It reaches ROC-AUC 0.871 on held-out CT and 0.919 on in-sample MRI, improving over organ-identity baselines by 0.110 and 0.104 AUC points. We vary the IoU threshold, predict IoU directly, test transfer to withheld anatomical families, and report calibration and conformal intervals. Our contributions are:

- quantifying the reliability gap on two modalities, including the silent failure mode that volume-based verification clears;
- a reference-free reliability predictor, with modality-dependent descriptor signals (fragmentation on CT and size on MRI);
- evidence beyond organ identity through organ-rate baselines and transfer to withheld anatomical families, while showing that per-structure holdout overstates transfer;
- calibrated and conformalized uncertainty estimates, with calibration reported where sample size supports it;

2. Related work

2.1. Automatic segmentation and reference-free quality control

Encoder-decoder networks such as U-Net (Ronneberger et al., 2015) and nnU-Net (Isensee et al., 2021) are foundational biomedical segmentation methods. TotalSegmentator (Wasserthal et al., 2023a) extended this paradigm to whole-body anatomy, training on 1,204 clinical CTs spanning scanners, contrast phases, and reconstruction kernels. It was later extended to 117 CT structures (Wasserthal et al., 2023b) and 50 structures on MRI (Akinici D’Antonoli et al., 2025). Because the distributed package exposes predicted masks but not logits or model internals, reference-free quality assessment must rely on the prediction, the image, or externally observable model behavior.

Prior approaches have extracted features from a predicted segmentation and regressed overlap against an unseen reference (Kohlberger et al., 2012). Reverse classification accuracy (Valindria et al., 2017), demonstrated at biobank scale on cardiac MR (Robinson et al., 2019), trains a classifier on the predicted

mask and scores it against an atlas, requiring per-image registration. Convolutional quality regression (Robinson et al., 2018) predicts Dice from the image and mask using a second network, while uncertainty-based methods use ensembles or dropout (Lakshminarayanan et al., 2017; Gal and Ghahramani, 2016), requiring access to model behavior. These methods have primarily addressed individual targets such as cardiac or brain structures and commonly require registration, an auxiliary network, or segmenter internals. We build on the feature-based approach (Kohlberger et al., 2012) by using lightweight descriptors computed from the mask and image alone, and evaluate them across whole-body anatomy, CT and MRI, held-out anatomical families, feature ablations, and uncertainty estimation.

2.2. Overlap metrics and quality thresholds

Quality is usually reported with Dice and IoU (Dice, 1945; Jaccard, 1912), related by $\text{IoU} = \text{Dice} / (2 - \text{Dice})$. Because both are volume ratios, a fixed threshold imposes a boundary tolerance that scales with structure size and penalizes small structures more severely for the same absolute error. The Metrics Reloaded framework (Maier-Hein et al., 2024; Reinke et al., 2024) argues that metrics should be matched to the task, while normalized surface distance (Nikolov et al., 2021) makes the tolerance an explicit per-organ quantity in millimetres.

Dice and IoU nevertheless remain the operational standards by which segmentation tools are commonly validated, and quality-control methods often binarize them at a fixed cutoff. We did not identify prior work systematically examining how this binarization interacts with structure size and voxel spacing or how it influences which descriptors appear predictive. We therefore retain IoU, vary its threshold, and use direct IoU regression to avoid committing to a single operating point.

2.3. Calibration, conformal prediction, and selective inference

Post-hoc calibration with Platt scaling (Platt, 2000) or isotonic regression (Zadrozny and Elkan, 2002) adjusts predicted probabilities without changing their ranking. Expected calibration error (Naeini et al., 2015; Guo et al., 2017) and the Brier decomposition into reliability, resolution, and uncertainty (Murphy, 1973) are standard diagnostics, but calibration itself

provides no finite-sample coverage guarantee. Conformal prediction can provide finite-sample marginal coverage under exchangeability (Vovk et al., 2005; Angelopoulos and Bates, 2023); conformalized quantile regression produces adaptive intervals (Romano et al., 2019), and Mondrian variants apply corrections to subgroups. Selective prediction frames the coverage–error tradeoff as a safety budget (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017).

Conformal methods have recently been applied to segmentation quality control. Wundram et al. (Wundram et al., 2024) derive heuristic Dice ranges from sampling-based uncertainty, compare five sampling strategies, and conformalize the ranges with split conformal prediction for a single retinal-vessel task. That approach presupposes the ability to draw many segmentation samples from the model and addresses one task and structure. To our knowledge, group-conditional coverage across anatomical families and the sample-size regime in which such conditioning is feasible have not been characterized for reference-free whole-body segmentation quality assessment.

3. Methods

3.1. Segmentation pipelines and data

For CT we use TotalSegmentator version 2 (task `total`, 117 structures) and its public v2.0.1 dataset (Wasserthal, 2023): 1,228 subjects at 1.5 mm isotropic resolution, partitioned 1,082/57/89 into train/validation/test. All CT results use the 57-subject validation partition held out from segmenter training; the official test partition is untouched. This measures cross-subject performance within the held-out validation distribution, not external generalization. The validation subjects yield 4,033 pairs; 16 empty predictions are flagged by an explicit zero-mask rule and excluded, leaving 4,017 non-empty masks, 74.5% of which meet the quality criterion.

For MRI we use the companion model (task `total_mr`, 50 structures) and public dataset (Wasserthal and Akinci D’Antonoli, 2025), with native, subject-variable spacing and train (190)/test (49) partitions but no validation split. We carve a subject-grouped 20% development subset (38 subjects) from the segmenter’s training partition, obtaining 550 pairs, 2 empty predictions, 548 masks, and a 23.0% accept rate. Because the segmenter saw this source partition, MRI is an exploratory in-sample analysis; the frozen test accept rate (16.9%) indicates shift but does not

bound optimism. For both modalities, images, predictions, and references are reoriented to canonical RAS, original axis codes are recorded, and reference masks under 20 voxels are treated as absent.

3.2. Reliability descriptors

Each mask yields nineteen descriptors from the mask and source image alone (Table 1); none needs a reference, the segmenter’s internals, or registration, and the same set is used on both modalities.

Relative position and scan-axis extent inherit field-of-view variation. CT intensities reflect both contrast and non-contrast studies. MRI intensity is not calibrated, so its five intensity descriptors are not comparable across subjects; MRI analyses are run with and without this group, which is flagged as unreliable. Reference-derived quantities (IoU, Dice, confusion counts, reference volume, and match status) build labels only and are excluded by assertion.

3.3. Labels and the threshold question

The primary label is accept when $\text{IoU} \geq \tau$ with $\tau = 0.90$ (Dice ≥ 0.947). Because overlap metrics are volume ratios, a fixed τ encodes a size-dependent standard: for a sphere of radius r with uniform boundary error δ ,

$$\text{IoU} = (1 - \delta/r)^3 \approx 1 - 3\delta/r,$$

so tolerated error grows linearly with r . With median reference volumes from our CT validation subjects, $\tau = 0.90$ tolerates 2.3 mm for the liver (1,390 mL) but 0.35 mm for the adrenal gland (4.6 mL), or 1.5 versus 0.23 voxels; recomputed from MRI’s coarser, subject-variable spacing, the adrenal tolerance falls to roughly 0.12 voxels, a physically unattainable standard that partly explains the low MRI accept rate. We therefore treat τ as a variable, repeating CT classification for $\tau \in \{0.70, 0.75, 0.80, 0.85, 0.90, 0.95\}$ and complementing classification on both modalities with direct IoU regression from the same descriptors. A secondary, clinically interpretable label is relative volume error $|n_{\text{pred}} - n_{\text{ref}}|/n_{\text{ref}}$ in voxel counts (the voxel-volume factor cancels), thresholded at 10% and 20%; its disagreement with the overlap label quantifies the failure mode invisible to volumetric checks.

3.4. Models and validation protocol

Models take $x \in \mathbb{R}^{19}$. We compare logistic regression, a depth-3 tree, a random forest (400 trees, minimum

Table 1: Reliability descriptors computed per predicted mask on both modalities. All are reference-free; ground-truth-derived quantities are excluded from the descriptor set by a programmatic assertion.

Group	Descriptor	Definition	Rationale
Size (3)	<code>num_voxels</code> , <code>volume_mm3</code> , <code>bbox_volume_mm3</code>	voxel count; physical volume; bounding-box volume	gross mis-sizing indicates failure
Position (3)	<code>centroid_{x,y,z}_rel</code>	mean coordinate over image size, per axis	anatomy occupies stereo-typed positions
Extent (4)	<code>bbox_{x,y,z}_rel</code> , <code>mask_to_bbox_ratio</code>	box extent per axis; mask voxels over box voxels	shape envelope and compactness
Connectivity (2)	<code>num_components</code> , <code>largest_component_fraction</code>	26-connected components; largest component over total	structures are singly connected; fragmentation signals failure
Boundary (2)	<code>touches_boundary</code> , <code>boundary_fraction</code>	any voxel within 3 of the edge; fraction on outermost face	field-of-view truncation and its severity
Intensity (5)	<code>mean/median/std_HU</code> , <code>p05/p95_HU</code>	intensity statistics within the mask (Hounsfield units on CT; arbitrary on MRI)	tissue-characteristic ranges; deviation suggests wrong tissue

leaf 2, balanced class weights), and histogram gradient boosting; all selection uses only training subjects within each outer fold. The raw-descriptor random forest is primary for CT classification and the IoU regressor for MRI. Because MRI acceptance at $\tau = 0.90$ is highly imbalanced, MRI ablations also use the relative-volume- error label and are reported separately from CT IoU-label ablations. We report minority average precision, with reject as the CT minority and accept as the MRI minority. Always-accept and organ-rate baselines (training-fold accept rate from organ identity) test whether models assess masks rather than memorize organ difficulty. Five-fold subject-grouped GroupKFold ensures all metrics are out-of-fold. Ablations report pooled and mean within-organ AUC for 66 CT structures with adequate balance (≥ 25 masks, ≥ 5 minority); the latter distinguishes structure difficulty from mask quality. CT transfer uses leave-one-structure-out and leave-one-family-out over 13 families. Since 24 near-identical ribs make the former interpolation, family holdout is stricter. Calibration uses separate 15-CT/12-MRI family groupings, distinct from transfer. MRI is an in-sample probe, so CT depth, threshold-sweep, and transfer analyses are not repeated there.

3.5. Calibration and conformal intervals

Probabilities use Platt scaling or isotonic regression in a nested protocol: an inner subject-grouped

split of each outer training fold fits the calibrator. We assess equal-count ECE and the Brier decomposition. For regression, conformalized quantile regression uses subject-disjoint calibration splits and $E_i = \max(q_{lo}(x_i) - y_i, y_i - q_{hi}(x_i))$; the $\lceil (n+1)(1-\alpha) \rceil$ -th score widens intervals symmetrically. A Mondrian variant computes corrections per anatomical family when feasible (≥ 4 calibration masks for 80% and ≥ 9 for 90%); otherwise it uses the global correction. Intervals are clipped to $[0, 1]$. Correlation among masks within a subject means exchangeability, and thus coverage, is only approximate and marginal over masks.

4. Results

4.1. Failure rates and silent volume errors

Of 4,017 CT masks, 25.5% fall below IoU 0.90, while the per-structure distribution is bimodal (Figure 1): 32 of 108 structures with adequate support pass above 90%, but portal and splenic veins (2.7%), costal cartilages (16.3%), and duodenum (17.1%) nearly always fail. MRI acceptance is 23.0%; no adequately supported structure clears 90%, although the liver has median IoU 0.93 and the worst structures are near-misses (0.76–0.83). Among overlap failures, 52.8% of CT and 69.5% of MRI masks have relative volume error below 10%, so a volume check clears most spatially wrong masks.

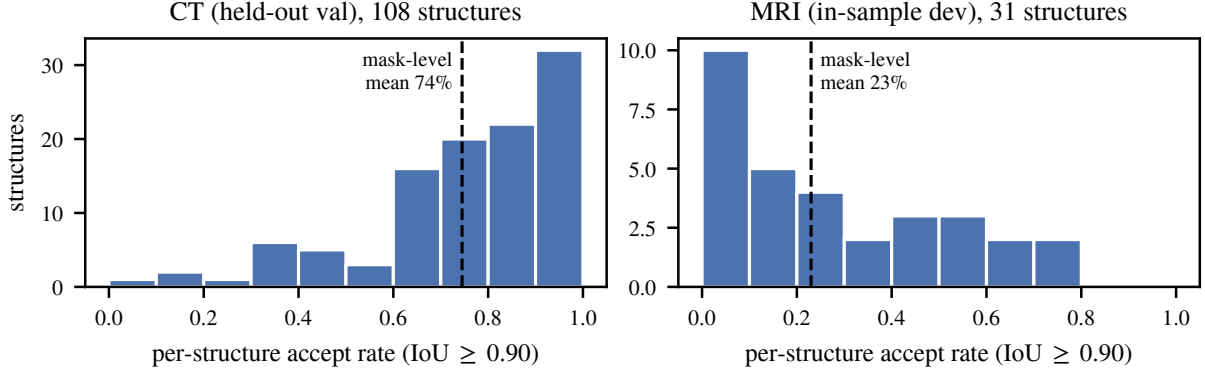


Figure 1: Per-structure accept rates at IoU 0.90 for structures with at least 20 (CT) or 8 (MRI) masks. The CT distribution is bimodal, so the mask-level mean describes few structures; on MRI no structure with adequate support exceeds 90%.

4.2. Reference-free reliability prediction

Table 2 summarizes the primary CT random forest and MRI IoU regressor at $\tau = 0.90$. Performance exceeds the organ-rate baseline by 0.110 and 0.104 AUC points, respectively. Adding organ identity to CT changes AUC only from 0.871 to 0.873. In the separate MRI classification analysis, dropping uncalibrated intensity descriptors marginally improves AUC (0.915 versus 0.910), so both variants are reported.

4.3. Descriptor behavior differs between CT and MRI

Feature-group ablation (Table 3) differs by modality. On CT, removing connectivity decreases AUC by 0.054 under the IoU label, 0.079 under the volume label, and 0.128 within organs; the within-organ result indicates a signal beyond structure difficulty. Removing size costs at most 0.010. On MRI, size has the largest effect (-0.050 ; -0.043 without intensity), whereas connectivity changes AUC by only -0.006 . MRI contains whole organs rather than CT’s subdivided ribs and vessels, so fragmentation is uncommon.

4.4. Threshold sensitivity

Sweeping the CT threshold from 0.70 to 0.95 leaves discrimination stable (Figure 2): classifier AUC stays within 0.867–0.877 while acceptance changes from 92.7% to 46.0%. A single IoU regressor (Spearman 0.693, MAE 0.064) is slightly stronger at $\tau = 0.70$

(0.886 versus 0.877), 0.014 behind at 0.90 (0.856 versus 0.870), and 0.037 behind at 0.95. On MRI, the accept prevalence falls to 3.3% at $\tau = 0.95$ and AUC rises from 0.887 to 0.978; this reflects a changing, increasingly rare target rather than improved prediction. At $\tau = 0.90$, the regressor reaches 0.919 (Spearman 0.81, R^2 0.60), subject to the in-sample caveat.

4.5. Transfer to unseen anatomy

Leave-one-structure-out over 66 CT structures gives mean AUC 0.821 (median 0.873), but 22 are ribs and transfer at 0.897 versus 0.782 for the others; holding out one rib leaves 23 near-identical siblings in training. Family holdout removes this redundancy (Table 4), giving mean AUC 0.779 across 13 families, with 12 above chance and lung at 0.930. Costal cartilage falls below chance (0.399), likely because its natural fragmentation inverts the connectivity signal and its sample is small ($n = 49$, 16% accepted). Superior vena cava (0.487) and brachiocephalic trunk (0.449) are also near chance. Above-chance prediction for 12 families supports a signal beyond organ identity, but does not rule out dataset-specific effects.

4.6. Calibration and prediction intervals

Table 5 and Figure 3 summarize calibration and conformal results. On CT, isotonic regression reduces ECE from 0.029 to 0.013 without a meaningful discrimination cost; the Brier reliability term is 0.0003 versus

Table 2: Primary-model performance against the organ-identity baseline. CIs are 95% subject-bootstrap intervals (1,000 draws). The minority class differs by modality, so average precision is reported for that class.

Modality	Masks	Accept	Model AUC [95% CI]	Organ-rate baseline	Minority AP
CT (held-out val)	4,017	74.5%	0.871 [0.852, 0.892]	0.761 [0.737, 0.785]	0.731 (reject)
MRI (in-sample dev)	548	23.0%	0.919 [0.890, 0.947]	0.815 [0.766, 0.861]	0.782 (accept)

Table 3: Change in ROC-AUC when a descriptor group is removed (leave-one-group-out). CT columns report pooled and within-organ AUC under the IoU label and pooled AUC under the volume-error label; MRI columns use the volume-error label. Fragmentation dominates on CT, size on MRI.

Group removed	CT, IoU label		CT, volume	MRI, volume	MRI, volume
	pooled	within-organ	pooled	pooled	no intensity
Size	−0.010	−0.018	−0.010	−0.050	−0.043
Position	−0.004	+0.003	+0.005	−0.001	−0.000
Extent	−0.002	+0.001	−0.002	−0.010	−0.000
Connectivity	−0.054	−0.128	−0.079	−0.006	−0.003
Boundary	+0.000	+0.001	+0.000	−0.003	−0.002
Intensity	−0.008	−0.001	+0.003	+0.001	n/a

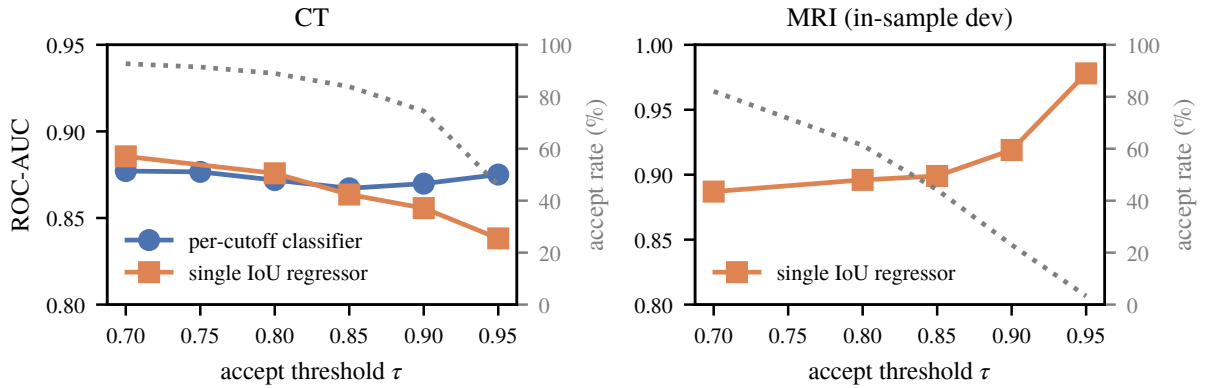


Figure 2: Threshold sensitivity. Left: on CT, per-cutoff classifiers and a single IoU regressor thresholded post hoc give near-identical discrimination across τ while the accept rate (dotted, right axis) halves. Right: on MRI the accept class becomes vanishingly rare at high τ and ROC-AUC drifts upward with prevalence rather than reflecting real gains.

resolution 0.0711, so performance is discrimination-limited. On MRI, neither calibrator improves the nested results, so probabilities are reported uncalibrated. Conformalization reaches near-nominal coverage on both modalities with roughly twice the MRI interval width, while width remains correlated with

error. Mondrian coverage is 81.8% on CT; 14 of 15 families meet the minimum of four calibration masks, but the reproductive family does not. On MRI, 10 of 12 families meet the floor but have at most 31 masks each; Mondrian lowers the width-error correlation from 0.45 to 0.36, so global intervals are reported.

Table 4: Leave-one-family-out transfer on CT: AUC of a model trained with the whole family withheld and evaluated on it. Twelve of thirteen families exceed chance; costal cartilage, whose natural fragmentation inverts the connectivity signal, is the exception.

Family	Masks	Accept	AUC	Family	Masks	Accept	AUC
Lung	233	0.880	0.930	Vertebrae	753	0.846	0.813
Muscle	313	0.879	0.884	Rib	939	0.765	0.793
Solid abdominal	153	0.699	0.861	Digestive	200	0.510	0.750
Heart	80	0.788	0.834	Airway and neck	77	0.675	0.723
Urinary	172	0.564	0.832	Great vessel	570	0.553	0.644
Shoulder bone	263	0.875	0.832	Costal cartilage	49	0.163	0.399
Neuro	85	0.835	0.829				

Table 5: Calibration (ECE, with AUC in parentheses) and conformalized interval coverage. Isotonic calibration helps on CT and fails on the small in-sample MRI subset; conformal coverage reaches nominal on both.

	CT	MRI (with intensity)	MRI (no intensity)
ECE, uncalibrated	0.029 (0.869)	0.036 (0.910)	0.041 (0.915)
ECE, isotonic	0.013 (0.866)	0.054 (0.881)	0.035 (0.899)
ECE, Platt	0.024 (0.868)	0.061 (0.883)	0.069 (0.890)
Coverage at 80%, raw intervals	52.0%	45.4%	n/a
Coverage at 80%, CQR (median width)	80.7% (0.091)	78.6% (0.158)	n/a
Coverage at 90%, CQR (median width)	90.7% (0.145)	87.0% (0.235)	n/a
Width-error Spearman, CQR	0.46	0.45	n/a

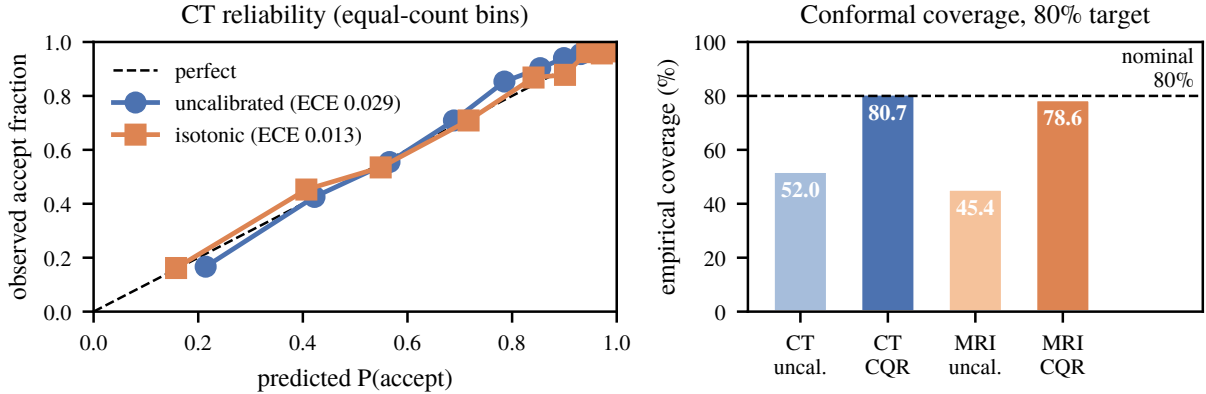


Figure 3: Left: CT reliability diagram before and after isotonic calibration (equal-count bins). Right: empirical coverage of the 80% interval before and after conformalization on both modalities; raw quantile intervals are badly overconfident and conformalization restores near-nominal coverage.

As a clinical illustration, we apply a published splenic-volume cutoff of 314.5 mL (Prassopoulos et al., 1997). No CT prediction changes the enlargement label, but one of 14 MRI masks does: a spleen measuring 322.9 mL in the reference is predicted as 312.8 mL. Its IoU is 0.858, while the reliability estimate is 0.908,

just above the default operating point. This is one illustrative case rather than a rate, but it shows how an unflagged mask can change a downstream clinical measurement.

5. Discussion

Nineteen mask/image descriptors predict per-mask quality beyond organ identity on CT and MRI, with evidence of transfer to withheld anatomy. The outputs include calibrated CT probabilities and conformalized intervals on both modalities, subject to approximate exchangeability within subjects. They could support review, abstention, or operating-point selection in automated clinical AI systems. The risk of an unguarded call is substantial: a quarter of held-out CT masks fail, most failures pass a volume check, and one MRI case shows a 3% volume error converting a true splenomegaly flag into a missed flag. Effects on clinical or agentic workflows remain untested.

Reliability does not transfer identically across modalities. Connectivity is most useful on CT, whereas size matters more on MRI; the fixed threshold tolerates 1.5 liver voxels on CT but roughly one eighth of an adrenal voxel on MRI; and calibration helps on 4,017 CT masks but not on 548 MRI masks. A reliability layer should therefore be validated separately for each modality and segmenter.

5.1. Limitations

MRI results are in-sample because the development subset is drawn from the segmenter’s training pool. The lower frozen-test accept rate (16.9% versus 23.0%) indicates shift but does not bound optimism; MRI findings are exploratory and require confirmation on an untouched partition. Both official test partitions remain unused. Family transfer includes one below-chance family, so atypical morphologies may require per-structure validation. The clinical-flip analysis is one illustrative organ/cutoff case, not a rate. IoU is retained as the operational standard despite its size bias; normalized surface distance (Nikolov et al., 2021) is the principled alternative for replication. We did not test the layer in a clinical or agentic workflow, and richer learned descriptors may improve its discrimination at the cost of the lightweight, black-box-compatible design.

6. Conclusion

Deployed segmentation tools return masks with no reliability signal, although a quarter of held-out CT masks and three quarters of in-sample MRI masks fail a conventional bar and most failures evade volume checks. A lightweight, reference-free layer predicts quality beyond organ identity, transfers to unseen CT anatomy, and provides calibrated CT probabilities and conformalized intervals on both modalities. These results support reporting reliability for each mask alongside aggregate segmentation accuracy when segmentation is used as a tool in clinical AI systems.

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